

The role of teachers' direct and emotional mentoring in shaping undergraduates' research aspirations: a social cognitive career theory perspective

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Abstract

Purpose – This study investigated the influence of teachers on undergraduate students' development of research aspirations and the mechanisms behind this process.

Design/methodology/approach – Employing social cognitive career theory, the study gathered data from 232 undergraduates, developed a structural equation model via the maximum likelihood method and executed empirical testing.

Findings – The findings reveal that neither direct nor emotional mentoring independently satisfies students' needs for self-efficacy and aspiration in research nor significantly influences research interest. Specifically, the study demonstrates that (1) research self-efficacy, outcome expectations and research interest significantly shape research aspirations; (2) an overemphasis on direct mentoring might impede research aspiration development and (3) a focus on emotional mentoring, while overlooking direct mentoring, could result in diminished research self-efficacy.

Originality/value – This research pioneers a comprehensive analysis of the role of teachers in shaping undergraduate research aspirations through the lens of social cognitive career theory. It underscores the critical need to both balance mentoring approaches and foster intrinsic research motivation.

Keywords Undergraduate mentorship, Social cognitive career theory, Mentor guidance, Research aspiration, Research self-efficacy, Mentoring relationships

Paper type Research paper

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Introduction

The global rise in undergraduate education and overall education levels underscores the growing importance of undergraduate involvement in scientific research. Research has consistently demonstrated that such participation offers valuable learning experiences (Seymour *et al.*, 2004). A pivotal element in this context is the undergraduate mentoring system, wherein university educators and professionals guide students across academic, career, and psychological aspects, fostering comprehensive development (Jung, 2018; Karaferye, 2018). In this study, the mentors are academic advisors assigned to undergraduate students in China, akin to graduate thesis advisors, responsible for guiding students in competitions, research projects, academic progress, and especially in degree thesis writing. Recognized for enhancing educational quality, stimulating learning motivation, and fostering innovative capabilities, this system plays a crucial role in shaping research aspirations (Leavitt *et al.*, 2022).

However, the efficacy of current undergraduate mentoring systems, particularly in China, faces critique due to over-reliance on personal relationships and an emphasis on mentor authority, potentially stifling innovation (Shi, 2016; Zhou *et al.*, 2019). Most existing studies on mentoring systems focus on systemic design, often overlooking the intricacies of mentor–student interactions and their impact on research aspirations.

The mechanisms through which mentors influence student research aspirations require clearer understanding. Pfund *et al.* (2006) highlighted the vital role of mentorship in enhancing research quality and student growth, yet there is a lack of comprehensive theoretical and empirical guidance for effective mentoring strategies aimed at boosting research aspirations. Moreover, the interaction between mentorship and student motivation for research remains underexplored. For instance, Overall *et al.* (2011) observed a complex, conditional mentor–student relationship, suggesting that fostering autonomy and setting clear expectations are key to effective mentorship.

Contrastingly, mentoring at the undergraduate level is an emerging area of study, with previous research primarily centered on graduate-level research behavior. For example, McDaniel *et al.* (2022) showed significant enhancements in doctoral students' research autonomy under effective mentoring systems. This highlights a gap in understanding the specific mentoring needs and mechanisms at the undergraduate level.

In summary, this study explores how mentors influence undergraduate students' research aspirations. We use social cognitive career theory (SCCT) as a theoretical framework and examine the effects of mentors' direct research mentoring and emotional mentoring on undergraduate students' research self-efficacy, research interests, and research aspirations. This study targeted undergraduate students and used structural equation modeling to analyze the data, aiming to understand the factors and mechanisms that affect students' willingness to participate in research from both theoretical and empirical perspectives. To enhance the stimulation of mentoring and cultivate students' research interests and aspirations, we propose suggestions and strategies with reference to value and enlightening significance.

This research focuses on the intricate dynamics of mentorship in undergraduate research, seeking to elucidate its multifaceted impacts. The study centers on key questions to understand the mentorship process better:

What is the influence of direct research mentoring by mentors on undergraduates' research aspirations, self-efficacy, and interests?

How does emotional mentoring by mentors shape undergraduates' research aspirations, self-efficacy, and interests?

In what ways can mentorship be optimized to bolster undergraduates' research aspirations and encourage their active participation in research?

When applied to student research, do the principles of self-efficacy and outcome expectation within the SCCT continue to hold a significant influence on students' research aspirations?

The study's primary contributions lie in reevaluating the commonly held belief that mentorship unilaterally enhances student research motivation. It enriches our understanding of the situational factors within SCCT, confirming its relevance in social sciences and specifically in the context of Chinese culture. By providing empirical insights into undergraduate research practices in China, this research offers valuable recommendations for enhancing mentorship approaches in Chinese universities, aiming to amplify the research competencies and aspirations of undergraduate students.

Literature review

The origin and mechanism of mentorship

The inception of the undergraduate mentorship framework is anchored in the collaborative research endeavors between faculty and students. This partnership, characterized by educators' deep engagement with students in both academic and extracurricular settings, contributes to a conducive atmosphere that positively influences students' scholarly achievements and career progression (Davis and Jones, 2020). The mentorship program in undergraduate education plays a supportive role in nurturing students' research skills and encouraging their pursuit of advanced academic qualifications (Eagan *et al.*, 2013). Crucial to this evolution is the recognition of diverse mentor backgrounds, experiences, and mentoring dispositions, which profoundly impact the mentor–mentee dynamic and, subsequently, the students' research directions and aspirations (Pfund *et al.*, 2022).

Different mentorship styles, such as direct research guidance and emotional support, are crucial in shaping students' self-efficacy, metacognition, and research intentions (Medakovich, 2019). Direct research guidance includes mentors providing hands-on assistance with research activities, expertise in research design, methodology, and data analysis, and guidance through academic publishing complexities, helping students develop essential critical thinking and technical skills (Leavitt *et al.*, 2022). Emotional support involves mentors offering encouragement, building confidence, and providing a safe space for discussing academic and personal challenges (Hudson, 2013).

The influencing factors of undergraduate research aspiration

Research aspiration among undergraduates, a reflection of their inherent motivation and desire to engage in scholarly research, is governed by a blend of internal and external forces. Central to these forces, within the ambit of SCCT, are internal factors which include individual attributes – demographics such as gender, age, and education level – and psychological constructs encompassing interest and outcome expectations (Butz *et al.*, 2018). These components not only forge research aspirations but also shape students' propensity for research participation (Lopatto, 2004; Mogk, 1993). Externally, mentorship emerges as a pivotal element, influencing students' trajectories toward graduate education and scientific careers. This influence was initially identified by Deemer *et al.* (2014) and subsequently expanded upon in the later works of Deemer *et al.* (2020), underscoring its significance within the context of academic and professional development. This notion of mentorship as a crucial environmental influence or source of social support aligns with the SCCT framework and echoes the principles of self-determination theory, underscoring the salience of external

The evolution of social cognitive career theory

SCCT, proposed by Lent and colleagues in 1994, elucidates the interplay between academic interests and career choices and performance (Lent *et al.*, 1994). Anchored in Bandura's social cognitive framework, the SCCT synthesizes principles of social cognition with empirical findings on academic self-efficacy (Lent *et al.*, 2005). SCCT posits that self-efficacy beliefs, when intertwined with anticipated outcomes, play a pivotal role not only in directing students' educational and vocational decisions but also in shaping the genesis of interests, which in turn profoundly influences their career paths and performance (Wu *et al.*, 2022). This study focuses on external factors, specifically the direct and emotional guidance received by students. According to Lent and Brown (2019), SCCT suggests that external influences, such as mentorship and social support, enhance self-efficacy and outcome expectations, subsequently shaping students' interests, goals, and actions in their academic and career development (Lent and Brown, 2019). Mentorship, in particular, is crucial for providing academic counsel and emotional encouragement, essential for educational and professional advancement (Mendez *et al.*, 2017; Young *et al.*, 2020).

Research model and hypotheses

Research hypotheses

Research self-efficacy. This study delves into the concept of research self-efficacy, defined as the confidence level of undergraduate students in their ability to successfully engage in academic research activities. Originating from the seminal work of Bandura, a renowned psychologist and the progenitor of social learning theory, self-efficacy is conceptualized as an individual's confidence in navigating challenges across various domains, including academic, social, and health-related spheres. In the realm of academia and scientific inquiry, research self-efficacy emerges as a critical predictor of success, underscoring the importance of this social cognitive concept in the academic and scientific research field (Livinți *et al.*, 2021). The study's scale for assessing research self-efficacy is based on, and linguistically refined from, the instrument developed by Bieschke *et al.* (1996).

Research interest. In educational research, interest is often explored as a mediating factor between self-efficacy and career intentions within the framework of SCCT. This concept, strongly influenced by individual learning experiences, has been consistently validated in prior studies (Lent *et al.*, 1994). Building on this, the current study adapts the SCCT model's concept of career interest to examine research interest in undergraduates. It is noteworthy that, unlike doctoral students who often require emotional rapport with their supervisors, undergraduates, due to their relative lack of research experience, are more reliant on direct guidance from their mentors (Kim *et al.*, 2018). This dynamic suggests that intimacy might play a lesser role in shaping their self-efficacy, and they might benefit more significantly from direct instruction and feedback from their supervisors (Zimmerman *et al.*, 1992). In this context, research interest is conceptualized as the individual's intrinsic curiosity and strong aspiration to engage in and further scientific activities.

Outcome expectation. Outcome expectation, a concept integral to understanding human behavior, is defined as the belief regarding the consequences or rewards ensuing from participation in specific activities or attainment of particular goals (Lent *et al.*, 1994). Golub *et al.* (2009) described it as a mental state of preparedness for outcomes, influencing individuals' emotional responses in task completion. Further exploring this concept, some scholars have investigated the impact of self-efficacy on interest development and

performance through experimental studies. Notably, they discovered that self-efficacy not only elevates the level of interest but also enhances its stability and persistence (Silvia, 2003). This relationship implies a positive correlation between self-efficacy and interest, a notion supported in the realm of research. For instance, Sawitri and Creed (2021) applied SCCT to analyze the interplay between self-efficacy, outcome expectation, and research interest. Their findings elucidated that research self-efficacy and outcome expectation can indirectly foster the development of research interest via research motivation.

Research aspiration. Research aspiration, a concept integral to understanding academic motivation, is defined as the intrinsic motivation or desire of undergraduates to engage in research projects. This concept is rooted in the principles of SCCT, which itself is an extension of Bandura's social cognitive theory. SCCT conceptualizes aspiration as the determination to undertake specific activities or achieve future outcomes, with a focus on the selection of career goals (Bandura, 1986). This study adapts this framework to examine the research aspirations of undergraduate students.

Within the SCCT framework, a critical interplay exists between self-efficacy, outcome expectation, and goal systems. When applied to goal selection, both self-efficacy and outcome expectation are seen to directly influence the choice of goals (Bandura, 1986; Lent *et al.*, 1994; Maddux *et al.*, 1982). Byars-Winston and Rogers (2019) research, employing structural equation modeling, indicated varying impacts of self-efficacy on research aspiration across different racial and ethnic groups, particularly in career and STEM fields. This variability underscores the need for further exploration into how these dynamics translate into the selection of research aspirations. Consequently, this study proposes the following hypotheses:

H1. Self-efficacy positively affects research aspiration.

H2. Outcome expectation positively affects research aspiration.

Furthermore, Lent's experimental findings suggest that self-efficacy indirectly influences goal selection through interest (Lent *et al.*, 1994). While these findings were initially applied to career choice, the applicability of this relationship in more nuanced academic contexts warrants further investigation. Therefore, this study also hypothesizes:

H3. Research interest positively affects research aspiration.

Mentor emotional mentoring and mentor direct mentoring. Within the framework of SCCT, contextual factors encompass a range of external influences that can either facilitate or impede individual agency, including family, societal, cultural, economic, and political elements. These factors, while impacting self-efficacy and interest, also interact with personal goals and behaviors. Yet, the specifics of these mechanisms remain underexplored (Lent *et al.*, 2005).

Mentoring, in the context of student research aspirations, is an influential environmental factor (Byars-Winston and Rogers, 2019). Easley and Wang (2017) highlighted the mentor's role as a critical social situational factor influencing student entrepreneurial behavior, even substituting parental influence in some cases. This study integrates mentorship – both direct research mentoring and emotional mentoring – into the SCCT's contextual factors, assessing their impact on student research aspirations.

Research on direct mentoring, such as Lopatto's (2007) survey of 2021 undergraduates, underscores its significance in enhancing the undergraduate research experience and fostering research interest. Similarly, Overall *et al.* (2011) found that academic mentors, through resource provision and process planning, can bolster students' research self-efficacy.

Regarding emotional mentoring, studies like [Armyanti's \(2020\)](#) emphasize mentors' role in boosting students' learning motivation and fostering independence. [Haeger and Fresquez's \(2016\)](#) research at California State University Monterey Bay further suggests the equal importance of emotional and cultural mentoring in developing research competencies.

However, mentorship as an external factor also has potential drawbacks. Excessive direct mentoring, detached mentor–student relationships, and untrained mentors might diminish its effectiveness or negatively impact research self-efficacy and interest ([Overall *et al.*, 2011](#)).

Drawing from these findings, this study utilizes the SCCT model to probe into the mentor's influence as a contextual factor on students' research self-efficacy, interest, and aspiration. To address the research gap in this domain, the study posits the following hypotheses:

Impact of direct research mentoring

- H4.* Mentor direct research mentoring positively impacts students' research self-efficacy.
- H5.* Mentor direct research mentoring positively impacts students' research outcome expectation.
- H6.* Mentor direct research mentoring positively impacts students' research interest.

Impact of emotional mentoring

- H7.* Mentor emotional mentoring positively impacts students' research self-efficacy.
- H8.* Mentor emotional mentoring positively impacts students' research outcome expectation.
- H9.* Mentor emotional mentoring positively impacts students' research interest.

Conceptual model

This study constructs a conceptual model grounded in SCCT to investigate the influence of both direct and emotional mentoring on undergraduate research endeavors, as illustrated in [Figure 1](#). The model integrates four key internal factors – research self-efficacy, outcome expectations, research interest, and research aspiration – along with two pivotal contextual factors: mentor direct mentoring and emotional mentoring. It posits that these mentoring types exert direct effects on students' research self-efficacy, interest, and aspiration. Concurrently, the model hypothesizes that students' research self-efficacy and outcome expectations indirectly influence their research interest and aspiration. Furthermore, it identifies research interest as a critical mediating variable that bridges the gap between mentorship styles and the internal factors influencing student researchers.

Questionnaire design

In refining the research methodology, this study developed a structured questionnaire using a 7-point Likert scale, guided by established survey methodology practices ([Nworie *et al.*, 2023](#)). This approach not only ensured accurate measurement of the constructs but also provided comprehensive data for structural equation modeling (SEM). The questionnaire

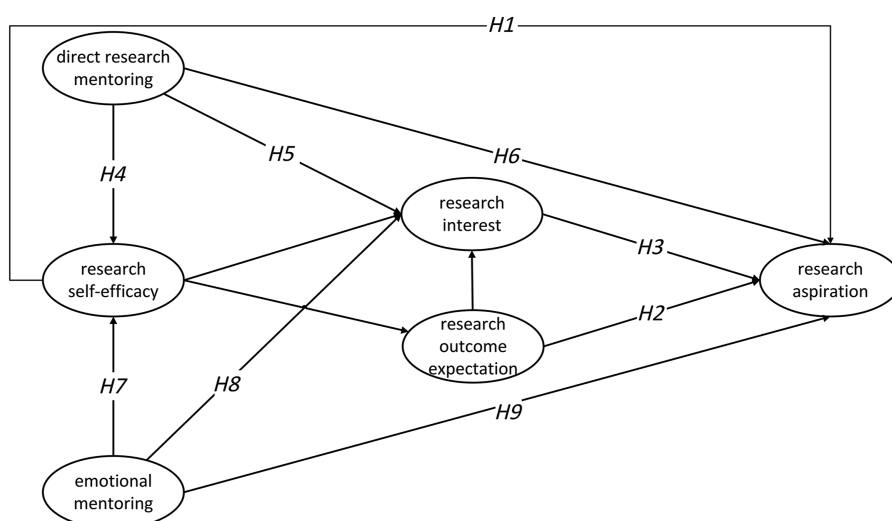


Figure 1.
Initial structural
equation model

Source(s): Created by author Xu Jun 2024-08-24

comprised 30 items: five items capturing basic demographic and academic information, 15 items focused on measuring aspects such as self-efficacy, outcome expectations, research interest, and research aspirations, and an additional 10 items specifically targeting direct research mentoring and emotional mentoring. This design was formulated to align with the SCCT framework. Survey participants were informed that in Chinese higher education, mentors are academic advisors guiding students in research projects, academic progress, and degree thesis writing throughout their undergraduate studies. The instrument's development integrated insights from validated scales (Bieschke *et al.*, 1996; Gibbons and Borders, 2010; Overall *et al.*, 2011; Vaccaro, 2009; Zhang, 2013) and incorporated cultural considerations for enhanced contextual relevance and integrating cultural factors; the specific content of the scales can be found in the appendix. A pilot test of 102 undergraduates showed Cronbach's alpha values between 0.871 and 0.939, exceeding 0.8 and indicating high reliability for formal distribution. The choice of a questionnaire-based method was driven by its proven efficacy in eliciting diverse and nuanced perspectives on the mentor–mentee relationship within the academic setting.

Data analysis

Data collection

To test the above hypotheses, this study conducted an online survey in 2023 through the Questionnaire Star platform, targeting students engaged in regular undergraduate education. A total of 242 valid questionnaires were collected, with an effective response rate of 88.6%. Following Höhne and Schlosser's (2017) suggestions for online surveys, questionnaires that took less than 80 s or more than 10 min to complete, or that showed serious inconsistency, were excluded. Regarding the sample size, the selection of 242 respondents was deemed appropriate for SEM analysis, especially considering the complexity of the model with 24 test items. This sample size aligns with recommendations by Chigwende and Govender (2020) for SEM, where they emphasize the importance of a sufficient sample size to achieve reliable model estimations in complex

analyses. Furthermore, the reliability of the survey data was ensured through various tests, including internal consistency checks and exploratory factor analysis, supporting the validity of the SEM approach employed in this study.

Regarding the demographic characteristics of the respondents, this paper conducted a survey based on the gender, discipline category, learning stage, and research experience of undergraduate students. The results are shown in [Table 1](#).

Data inspection

AMOS 24.0 software was selected for its sophisticated SEM capabilities, enabling robust maximum likelihood estimation and confirmatory factor analysis. This choice is supported by academic research, such as [Shi et al.'s \(2021\)](#) study, demonstrating its efficacy in handling complex models and ensuring the reliability of SEM results.

Reliability test. Reliability reflects the consistency or stability of questionnaire measurement results. This study extracted the path factor loadings obtained by the AMOS 24.0 measurement. There is no unified standard for its acceptable range, but relevant research suggests that higher than 0.5 is an acceptable standard, and higher than 0.7 is considered a clear structural goal, indicating that there is a very strong correlation between the observed variables and the latent variables ([Hair et al., 2017](#)). [Table 2](#) displays factor loadings for each survey item, all exceeding 0.75, affirming the robust reliability of the individual test items in our research instrument. This high level of reliability indicates that the measured constructs are consistently represented across the sample.

The composite reliability (CR) value and average variance extracted (AVE) value were calculated by SPSS. According to [Fornell and Larcker's \(1981\)](#) study, the CR value should be greater than 0.7 and the AVE value should be greater than 0.5. [Table 3](#) presents the AVE and CR values for each construct, demonstrating that all exceed the thresholds of 0.5 and 0.7, respectively. These metrics affirm the questionnaire's strong internal consistency and the reliability of our constructs, underpinning the soundness of our research findings.

Table 1.
Summary of
respondents'
demographic
characteristics

Attribute	Group	Number of participants	Percentage of total sample
Gender	Male	124	53.4%
	Female	108	46.6%
Academic level	Freshman	50	21.6%
	Sophomore	76	32.8%
	Junior	62	26.7%
	Senior or above	44	19.0%
Discipline category	Philosophy	5	2.2%
	Economics	30	12.9%
	Engineering	37	15.9%
	Science	33	14.2%
	Education	33	14.2%
	Management	24	10.3%
	Jurisprudence	12	5.2%
	Medical science	12	5.2%
	Agronomy	15	6.5%
	Art	10	4.3%
	Literature	16	6.9%
	History	5	2.2%
Research experience	Yes or have	157	67.7%
	No or without	75	32.3%
Source(s): Authors' own creation			

Variable	Measurement	Factor loading
Research self-efficacy	RS1	0.822
	RS2	0.826
	RS3	0.836
	RS4	0.770
Research outcome expectation	RO1	0.773
	RO2	0.792
	RO3	0.834
Research interest	RI1	0.771
	RI2	0.824
	RI3	0.778
	RI4	0.795
Research aspiration	RA1	0.750
	RA2	0.759
	RA3	0.870
	RA4	0.781
Direct research mentoring	DM1	0.784
	DM2	0.809
	DM3	0.839
	DM4	0.817
	DM5	0.796
Emotional mentoring	EM1	0.850
	EM2	0.817
	EM3	0.828
	EM4	0.814
	EM5	0.843

Source(s): Authors' own creation

Table 2.
Results of factor
analysis

Variable	Cronbach's alpha	CR	AVE
Research self-efficacy	0.837	0.887	0.662
Research outcome expectation	0.727	0.842	0.640
Research interest	0.832	0.871	0.628
Research aspiration	0.820	0.870	0.626
Research direct mentorship	0.885	0.905	0.655
Emotional mentoring	0.902	0.905	0.655

Source(s): Authors' own creation

Table 3.
Reliability and validity
analysis results: CR,
Cronbach's alpha,
and AVE

Validity test. For the validity judgment, there are the following criteria: if a variable's correlation coefficient with other variables is less than the square root of its AVE, it indicates that the variable has good discriminant validity (Fornell and Larcker, 1981). As demonstrated in Table 4, the square roots of the AVE values, placed diagonally, surpass inter-construct correlation coefficients, validating the discriminant validity of our constructs. This ensures that each construct is distinctly measured and sufficiently independent from others within our model.

HTMT test. In addition, this study used heterotrait-monotrait (HTMT) ratio to avoid potential multicollinearity problems in latent variables, which is the ratio of between-construct and within-construct correlations (Henseler et al., 2015), which may affect the estimation of path coefficients and model fit. In addition, HTMT testing can provide more accurate results than other traditional methods (such as Fornell-Larcker criterion or cross-

	Research self- efficacy	Research outcome expectation	Research interest	Research aspiration	Research direct mentorship	Emotional mentoring
Research self- efficacy	0.831					
Research outcome expectation	0.744	0.810				
Research interest	0.423	0.370	0.792			
Research aspiration	0.328	0.410	0.534	0.793		
Research direct mentorship	0.186	0.214	0.443	0.503	0.800	
Emotional mentoring	0.296	0.453	0.691	0.471	0.353	0.815

Table 4.
Discriminant validity

Source(s): Authors' own creation

loading), especially when the measurement model is homogeneous (Roemer *et al.*, 2021). Henseler *et al.*'s (2015) study showed that a HTMT value of 0.85 is a more conservative and strict threshold that can effectively detect discriminant validity problems. Table 5 elucidates that the highest HTMT value observed in our study is 0.831, aligning with stringent criteria for the HTMT assessment. This result underscores the model's reliability and the distinctiveness of the constructs, reinforcing the validity of our analytical framework.

Robustness test. Liu *et al.* (2021) verified the robustness of their model of facial recognition and privacy research in China by adding demographic control variables and comparing the changes in paths before and after. To test the robustness of the model in this study, we selected the grade, gender, major, institution, and past research experience of students as control variables and analyzed them using AMOS software. The robustness test outcomes in Table 6 indicate that after incorporating control variables, the path coefficients and fit

	Research self- efficacy	Research outcome expectation	Research interest	Research aspiration	Research direct mentorship	Emotional mentoring
Research self- efficacy						
Research outcome expectation	0.353					
Research interest	0.471	0.502				
Research aspiration	0.694	0.448	0.536			
Research direct mentorship	0.452	0.210	0.414	0.376		
Emotional mentoring	0.293	0.179	0.328	0.428	0.744	

Table 5.
Differential validity (HTMT)

Source(s): Authors' own creation

Path/indicator	Before added	After added
DM → RS	0.522***	0.522***
DM → RI	0.184	0.184
DM → RA	-0.246***	-0.239***
EM → RS	-0.089	-0.089
EM → RI	0.056	0.056
EM → RA	0.348***	0.346***
RS → RA	0.568***	0.563***
RO → RA	0.138**	0.140**
RI → RA	0.186**	0.193**
RSMEA	0.410	0.410
χ^2/df	1.395	1.393
CFI	0.972	0.961
TLI	0.968	0.957

Note(s): *** At the 0.01 level (double tail), the correlation is highly significant

** At the level 0.05 (double tail), the correlation is significant

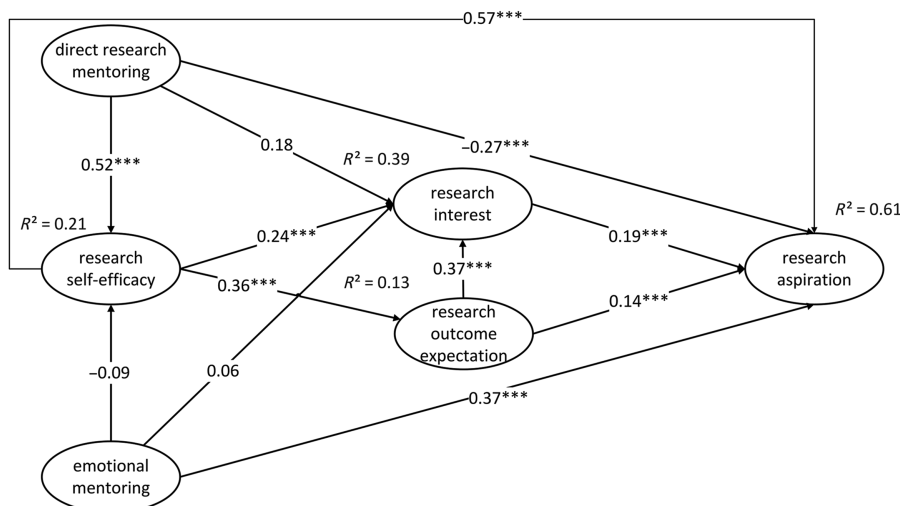
Source(s): Authors' own creation

Table 6.
Robustness test

indices remain substantially unchanged. This stability attests to the model's high resilience and reliability, suggesting that the identified relationships hold consistently across different contextual backgrounds.

Structural model

This study utilized the maximum likelihood estimation method to validate the structural model, delineating the explained variance (R^2 values) for each construct within our model through AMOS 24.0 software, as depicted in Figure 2. Specifically, the R^2 values for research



Note(s): **At the 0.05 level (double-tailed), the correlation is significant

***At the 0.01 level (double-tailed), the correlation is highly significant

Source(s): Created by author Xu Jun 2024-08-24

Figure 2.
Structural model

self-efficacy, outcome expectation, research interest, and research aspiration are 0.21, 0.13, 0.39, and 0.61, respectively, illustrating the substantial explanatory power of our model, particularly for research aspiration. The context of these R2 values suggests that our model effectively captures the key determinants influencing undergraduates' research aspirations and self-efficacy. Comparatively, these R2 figures align well with benchmarks from other studies applying SCCT in different contexts, such as [Lent et al. \(2005\)](#) and [Mohd Rasdi and Ahrari \(2020\)](#), thereby affirming our model's robustness. Notably, the variations in R2 for research self-efficacy and outcome expectation underscore the nuanced impact of individual and contextual factors within our theoretical framework, reinforcing the model's relevance and applicability in understanding the dynamics of research aspirations among undergraduates.

This study tested hypotheses [H1](#) to [H9](#) and found that [H1](#) to [H4](#) and [H9](#) were supported, while [H6](#) was contrary to expectations. Specifically, research self-efficacy ($\beta = 0.57$) and outcome expectation ($\beta = 0.14$) were significantly positively correlated with research aspiration ([H1–H2](#)), and research interest was also significantly positively correlated with research aspiration ($\beta = 0.19$) ([H3](#)).

For the situational perspective, direct research mentoring from mentors was significantly positively correlated with research self-efficacy ($\beta = 0.52$) ([H4](#)), but not related to research interest ([H5](#)), and was significantly negatively correlated with research aspiration ($\beta = -0.27$) ([H6](#)). Emotional mentoring from mentors was only significantly positively correlated with research aspiration ($\beta = 0.37$) ([H7](#)) but not related to research self-efficacy or research interest ([H8–H9](#)).

Discussion

Result discussion

The impact of internal factors on research aspiration. This investigation revealed a notable and substantial link between research self-efficacy, aspirations, and interests, substantiating the relevancy of SCCT's internal constructs within undergraduate research paradigms, predominantly concerning self-efficacy and outcome expectations. Notably, the influence of self-efficacy on research aspirations outstripped that of outcome expectations, likely attributable to self-efficacy's closer alignment with students' tangible behaviors. Conversely, outcome expectations might seem comparatively abstract or removed, potentially elucidating the observed direct and robust impact of self-efficacy on research aspirations within our SCCT-aligned structural model ([Lent et al., 2003](#)). This alignment underscores the imperative role of self-efficacy in fostering concrete research aspirations among undergraduates, as demonstrated in our findings.

The impact of direct mentoring on research aspiration and self-efficacy. This investigation delineates the nuanced influence of direct mentoring on undergraduates, identifying a nuanced dynamic where direct mentoring significantly augments research self-efficacy but, paradoxically, may attenuate research aspirations. These findings unfold through multiple dimensions.

Augmentation of self-efficacy: The study illustrates that direct mentoring enriches undergraduates' research self-efficacy via targeted feedback, methodical guidance, and demonstrative skill modeling, echoing findings from [Bieschke et al. \(1996\)](#) and [Chemers et al. \(2001\)](#). Such fortified self-efficacy potentially uplifts their research engagement, validated by established links to academic commitment ([Petko et al., 2020](#)).

Dampening of intrinsic motivation: Concurrently, our analysis suggests that while direct mentoring fosters competence, it might inadvertently diminish intrinsic motivation for research, a vital component elucidated by self-determination theory ([Ryan and Deci, 2000](#)).

Such a mentoring approach risks overlooking the essential elements of autonomy and connectivity, integral for nurturing enduring intrinsic motivation.

Risk of over-intervention: The findings further delineate risks associated with excessive mentor oversight, which could undermine students' autonomous drive, aligning with insights from [Pelletier et al. \(2002\)](#). This excessive guidance may inadvertently stifle long-term research engagement, manifesting a critical dichotomy within the mentoring paradigm.

The influence of emotional mentoring on research aspiration and research self-efficacy. This inquiry delineates the nuanced repercussions of emotional mentoring on undergraduates, spotlighting a substantial enhancement in research aspirations without a corresponding increase in self-efficacy. The bifurcated outcomes of this mentoring style are elaborated as follows.

Emotional support versus skill acquisition: Emotional mentoring's forte lies in sculpting a nurturing environment, akin to the solidarity observed in peer cohorts, which, while alleviating stress, falls short in fortifying undergraduates' self-assessment of their research competencies. This mentoring style, devoid of the direct skill transfer seen in direct mentoring, underscores the necessity for a balanced mentorship approach that integrates both emotional support and skill development to optimize research skill augmentation and passion ([Betz and Voyten, 1997](#); [Zimmerman et al., 1992](#)).

Challenges within the mentorship framework: Further scrutiny within this research reveals systemic hurdles embedded in the mentorship schema, like ill-defined goals and suboptimal student involvement ([Lambert et al., 2022](#)), that might attenuate the potency of emotional mentoring in bolstering self-efficacy.

Conversely, the pronounced enhancement of research aspirations through emotional mentoring underscores its indispensable role in engendering a conducive and engaging mentor–mentee nexus, pivotal for nurturing steadfast commitment and identity formation within the academic arena ([Hernandez et al., 2013](#); [Kram and Isabella, 1985](#)). This segment of mentorship emerges as a linchpin in sculpting students' professional identities and competencies, prerequisites for their enduring involvement in scholarly research ([Hudson, 2013](#)).

Assessment of mentorship's impact on research interest. This analysis delineates that neither direct nor emotional mentoring significantly alters undergraduates' research interest, reflecting its inherent stability as a deeply ingrained personal attribute. This study supports the idea that mentorship provides essential support but does not directly reshape the foundational aspects of research interest, which is consistent with [Deemer et al.'s \(2020\)](#) findings. Our investigation confirms that research self-efficacy is a crucial factor in maintaining research interest, which aligns with the views presented by [Renninger and Hidi \(2015\)](#). In alignment with the foundational principles of SCCT, the empirical evidence we garnered suggests that intrinsic determinants such as self-efficacy and outcome expectations predominantly orchestrate research interest. Concurrently, while mentorship contributes an ancillary and indirect role, it is the internal constructs that predominantly sculpt research interest, as posited by [Lent \(2004\)](#).

Revelation

Addressing challenges in China's undergraduate mentorship. In addressing the complexities of China's undergraduate mentorship system, significant challenges arise from resource constraints, cultural dynamics, systemic structures, and prevailing mindsets. Key issues identified include ineffective mentor–student communication, unrealistic mentor expectations, undue mentorship pressure, and students' diminished trust and respect towards mentors, coupled with their often vague or passive academic goals ([Zhou et al., 2019](#)). Navigating this landscape necessitates undergraduates to possess advanced cognitive

skills, robust emotional support, and a solid sense of research self-efficacy. Mentors, therefore, play pivotal roles, not only as academic facilitators providing guidance but also as emotional supporters enhancing student resilience (Medaille *et al.*, 2022). However, the research findings indicate that neither direct nor emotional mentoring alone suffices in meeting the comprehensive needs for students' research self-efficacy and aspirations.

In light of these insights, this study proposes several actionable recommendations:

- (1) Educational institutions should implement targeted mentor training programs, augment faculty resources, and guarantee substantial direct mentoring.
- (2) Mentors are encouraged to reevaluate and move beyond the preconception that nurturing undergraduates yields limited academic returns, instead fostering patience and encouragement to develop the potential academic leaders of tomorrow (Tsang, 2018).
- (3) There is a need for mentors to cultivate students' autonomy and sense of belonging within the academic community, which is instrumental in generating genuine motivation for research endeavors.

Balancing direct mentoring. The study's findings highlight the pivotal role of direct research mentoring in enhancing undergraduates' research self-efficacy. To optimize its benefits, several strategies are recommended:

- (1) Educational institutions should actively promote and value direct mentorship as a means to bolster students' research capabilities and confidence.
- (2) Mentors are advised to provide comprehensive guidance encompassing feedback on research methodologies, theoretical underpinnings, assistance in research planning and proposal development, and instruction in academic writing standards.

However, the study also indicates that excessive direct mentoring may inadvertently dampen students' research aspirations. In addressing this concern:

- (1) Educational institutions should establish clear guidelines and offer regular training to mentors, ensuring balanced involvement in student research projects.
- (2) Mentors should aim to achieve a delicate balance in their guidance, allowing sufficient intellectual space for students to engage in independent thought and exploration.

Emotional assistance cultivation. This study underscores the role of positive mentor–student relationships in fostering undergraduates' research engagement. To harness this potential, the following measures are advocated:

- (1) Educational institutions should formulate initiatives to nurture positive mentor–student dynamics, such as organizing seminars aimed at encouraging undergraduate research involvement.
- (2) Mentors should receive training to adopt a personalized approach for undergraduates, considering their backgrounds and specific needs. Balancing emotional support with academic guidance is crucial to avoid diminishing research self-efficacy, ensuring a distinct approach from that used with graduate students.
- (3) Providing ample research resources, alongside additional support such as mechanisms for course credit acquisition, can substantially lower students' opportunity costs and boost their research participation (Khoo, 2018).

- (4) Undergraduates should proactively engage in long-term research planning, recognizing the significance of their involvement, initiating projects independently, and considering mentors as facilitators rather than primary decision-makers in their research endeavors (Wang, 2019).

Limitations

The primary limitation identified in this study pertains to its sample concentration within South China, which potentially restricts the extrapolation of our results. Such geographical specificity raises concerns regarding the universality of our findings across disparate educational contexts, thus narrowing the scope of their interpretability. Moreover, the investigation's delineation of mentoring paradigms and the nuanced interplay between mentors and mentees – crucial for elucidating the mentorship's influence on students' research aspiration – lacks comprehensive detail. This omission restricts the breadth of our conclusions and fails to encompass the diversity of mentoring approaches documented within our sample. While the research offers substantive insights into the mentorship's role in bolstering undergraduate research motivation, its applicability remains confined to the investigated context, given the scant examination of mentor characteristics and practices. Future inquiries should endeavor to meticulously delineate these aspects, examining the spectrum of mentorship styles and their potential variations across educational landscapes. Augmenting the SCCT framework to incorporate a nuanced appraisal of mentor roles would significantly enrich our comprehension of mentorship efficacy and, in turn, fortify the external validity of this research, thereby amplifying its academic contribution to the educational domain.

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Appendix

Survey questionnaire

Part A: Demographic information

School: Discipline: Current Level of Education:

Participation status:

Have you ever had an academic advisor?

Have you had experience in academic conferences, paper publishing, research projects, or academic competitions?

Part B: Research self-efficacy, interests, outcome expectation and aspirations

Rate each statement on a scale from 1 (Extremely Inconsistent) to 7 (Extremely Consistent)

- (1) I am confident that I am competent for scientific research
- (2) I am confident that I can handle the difficulties in scientific research
- (3) I am good at scientific research analysis
- (4) I can express the research results very well
- (5) Willing to read research journal articles
- (6) Willing to be a member of the research team
- (7) Willing to lead the research team

- (8) Willing to develop data analysis strategies for research
- (9) Doing research will enhance my sense of self-worth
- (10) I believe that scientific research skills will be very helpful to my employment
- (11) I have a clear understanding of how to research
- (12) Hope to make achievements in professional fields
- (13) Hope to learn more data processing methods
- (14) Hope to further my study in my professional field
- (15) Hope to apply for scientific research project fund

Part C: Mentor support and relationship

Rate each statement on a scale from 1 (Extremely Inconsistent) to 7 (Extremely Consistent)

- (1) Clarify my expectations and goals in scientific research
- (2) Make institutional plans for me in scientific research
- (3) Provide me with pertinent suggestions on scientific research
- (4) Help me improve my scientific research writing
- (5) Teach me the knowledge and methods related to scientific research
- (6) When I encounter difficulties, he will help me
- (7) My mentor affirms my value and efforts when I am engaged in scientific research
- (8) When discussing my research with me, my mentor will be very enthusiastic and positive
- (9) Will listen kindly and respond to my concerns
- (10) Helped me and completed my scientific research task together

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